

Abhedya: The Next-Generation Web Security Engine

Created by: Shivang Kaushal & Sambit Kumar Garanayak  
Version: 1.0 (Edge Shield Beta)

Welcome to **Abhedya** (Sanskrit for *Impenetrable*).

Traditional antiviruses and web scanners rely on outdated text-matching and blacklists, leaving you vulnerable to "zero-day" phishing attacks—fake websites created today that haven't been reported yet. Abhedya changes the game. By combining lightning-fast heuristic checks with Google's powerful **Gemini 2.5 Flash** vision AI, Abhedya actually *looks* at the websites you visit, catching visual mismatches and deceptive intent before you can click or type.

The Shield in Action

(Developer Note: Take screenshots of your block screen and new control panel, place them in your project folder, and replace the placeholder text below with the actual image links!)

| The Block Screen                                | The Control Panel                       |
|-------------------------------------------------|-----------------------------------------|
|                                                 |                                         |
| Abhedya intercepting a zero-day brand mismatch. | The new Adaptive Threat Threshold menu. |

Why Choose Abhedya? (Core Features)

We built Abhedya to be ruthlessly effective against threats, but completely invisible when you are browsing safely.

- Multimodal "Human-Eye" AI:** Abhedya doesn't just read the website's code; it takes a split-second screenshot. If the hidden code says "Instagram" but the screen looks like "Google Gemini" (or a fake banking login), the AI catches the mismatch and drops the shield instantly.

-  **Zero-Tolerance NSFW Blocker:** Powered by robust AI safety filters, Abhedya instantly flags and blocks explicit adult content, pornography, and illegal gambling interfaces with an unyielding 100/100 threat score.
-  **Zero-Lag Memory Core:** Security shouldn't slow down your internet. Abhedya features a smart PostgreSQL caching system. If you (or anyone else) visits a site, the safety score is cached for 24 hours. The next time you open it, the scan takes exactly 0.0 seconds.
-  **The Tranco Pardon:** Annoyed by overactive security extensions? Abhedya integrates the official Tranco Top 1000 global domains list. Verified sites (like YouTube, Google, or Wikipedia) receive an automatic score pardon, guaranteeing zero false positives on your daily websites.
-  **Resource Optimized:** Abhedya waits until you are *actually looking* at a tab before it scans. Open 20 links in the background? Abhedya stays asleep until you click them, saving your CPU, RAM, and bandwidth.

### Adaptive Threat Thresholds

Not every network is equally safe. Abhedya features a sleek, interactive control panel right in your browser toolbar, allowing you to scale your security level on the fly:

-  **Paranoid Mode (Blocks > 35):** Maximum lockdown. Recommended for public Wi-Fi or when handling highly sensitive financial data. Blocks anything remotely suspicious.
-  **Guarded Mode (Blocks > 50):** High security. The perfect setting for exploring unknown parts of the web or clicking links from emails.
-  **Standard Mode (Blocks > 65):** The balanced, everyday shield. Let's the normal web through but aggressively catches phishing and typosquatting. *(Default)*
-  **Relaxed Mode (Blocks > 80):** Low friction. Only blocks guaranteed, 100/100 fatal threats (like explicit NSFW content or confirmed malware drops).

### How It Works (The 5-Pillar Ensemble Engine)

#### WEB SECURITY ENGINE

Under the hood, Abhedya calculates a master threat score (0-100) using a heavily weighted, military-grade matrix:

1. **URL Impersonation (15%):** Mathematically compares the web address against the Top 1000 global brands to catch typosquatting (e.g., instagram.com).
2. **SSL Chain Analysis (15%):** Inspects the site's security certificate. Penalizes certificates that were created less than 14 days ago (a classic sign of a "burner" phishing site).
3. **Domain Age (20%):** Executes a live WHOIS lookup. Sites registered less than 30 days ago are treated with extreme prejudice.

4. **DOM Obfuscation (20%):** Scans the raw HTML for hacker tricks—hidden elements (display: none), excessive scripts, or disguised password-stealing forms.
5. **Gemini AI Intent (30%):** The final judge. Cross-references the visual screenshot with the raw code to determine the true intent of the page. If the AI detects a critical threat, an **Instant Kill Switch** overrides the math and locks the block score to 100.

### Installation Instructions (Beta Testers)

Because this is an unreleased Beta, you will need to "sideload" the extension into Chrome. It takes 15 seconds:

1. Download and **extract/unzip** the Abhedya-Beta-v1.zip file to a permanent folder on your computer.
2. Open Google Chrome and type `chrome://extensions` into the URL bar and hit Enter.
3. In the top right corner, turn ON the toggle switch for **Developer mode**.
4. In the top left corner, click **Load unpacked**.
5. Select the Abhedya-Beta folder you extracted in Step 1. *(Make sure you select the folder containing manifest.json and index.html).*
6. **Done!** Pin the  Abhedya shield icon to your browser toolbar to access your control panel.

*Note: Every time you open Chrome, it might show a small popup saying "Disable developer mode extensions." Just click "Cancel" or the "X" to keep your shield active!*

### Privacy First

Abhedya protects you without spying on you. It only activates when you navigate to a new page. The extension securely transmits the data to our cloud engine for a split-second threat analysis and immediately discards it. We do not track your browsing history.

**ABHEDYA**  
WEB SECURITY ENGINE

